

Chapter 29: Optical and electrical properties of nanostructured CdS-PS: p-Si heterostructure

Samarjit Sarmah^{1*}

^{1*} Department of Physics, Pub Kamrup College, Baihata Chariali, Kamrup-781381, Assam, India
samarjitsarmah59@gmail.com

1. Introduction

Photoconductivity is a well-known phenomenon in semiconductor physics, which indulge conversion of light energy into electrical signal [1]. Photodetectors (PDs) and photovoltaic devices (PVs) rely on this basic principle. These devices have found many important applications in radiation detection and solar cell industries [2-4]. Furthermore, extensive research has been under way for developing heterostructure based PDs and PVs in the field of material science in recent years [5-8]. However, one of the biggest challenges for configuration of such heterostructure is the choice of appropriate material. Porous silicon (PS) is one of such material which attracts great interest due to its fascinating electrical and optoelectronic properties [9,10]. PS can be fabricated by methods like electrochemical anodization, metal assisted chemical etching [11] etc. PS fabricated on p-type Si (p-Si) wafer acts as active hole transport layer, thus by deposition of electron rich material on PS: p-Si substrate p-n type heterojunction can be developed. There are various electron transport materials such as ZnO, ZnS, TiO₂, CdS etc. [12, 13], which can be employed for devising UV-vis PDs. However, processing and deposition of such materials are quite complex as well as expensive [14]. Metal sulphides, in particular nano structured CdS is easy to process and can be grown on large scale. Also, as reported in literature, higher electron mobility of CdS ($\sim 350 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) as compared to some other metal oxides [15], makes it a good candidate for heterostructure configuration in device application. In a recent work by Ismail et al. [16], the reported parameters of the CdS-PS based PDs do not show much promising result. This is possibly because of the fact that in their method they adopted CdS deposition by drop casting on PS wafer. There is another work by Manna et al. [17] on CdS deposited Si nanowire made by anodization of Si wafer. Their method of sample preparation involves complex and expensive techniques. Present chapter aims at fabrication of nanostructured CdS-PS:p-Si MSM heterostructure by simpler method yet attaining much better result in terms of photoresponsivity.

2. Experimental

2.1 Preparation of Al/nano CdS-PS:p-Si MSM heterostructure

The preparation of PS:p-Si structure and CdS nanoparticles have already been described in detail in Chapter 2, under subsection 2.2.1 and 2.2.3 respectively. Synthesized CdS nano powder weighing 0.15 g is deposited simultaneously on PS:p-Si as well as on glass substrate using Hind high vacuum coating unit by thermal evaporation method in a vacuum condition of $\sim 10^{-5}$ mbar. The thickness of the deposited CdS nanoparticle layer is maintained at ~ 170 nm by quartz crystal digital thickness monitor (DTM)

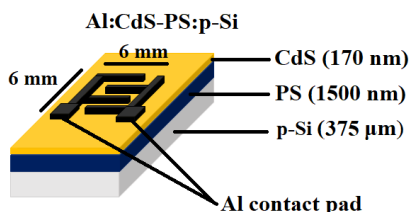


Fig. 1 Schematic of Al: CdS-PS: p-Si MSM heterostructured PD

attached to the chamber. Further construction of metal semiconductor metal (MSM) heterostructure is carried out in the same coating unit under the same condition. For this highly pure (99.99%) Al metal electrode of ~ 20 nm thickness is coated on CdS-PS:p-Si heterostructure and bare PS:p-Si (for comparison) structure using interdigitated shadow mask of dimension $6\text{ mm} \times 6\text{ mm}$. Schematic of the heterostructure is depicted in fig. 1. Thin n-CdS layer is preferred, so that equal number of charge carriers could be collected at the Al electrodes in equal time without recombination as e^- mobility at room temperature is reported to be greater (3 times) than h^+ mobility [18], which will consequently increase the photocurrent of the heterostructure.

3. Results and discussion

3.1 Morphological analysis

Morphology of the prepared CdS nanostructure is observed from FESEM and FETEM analyses, which are shown in fig. 2. Fig. 2(a) shows planar FESEM micrograph of the prepared nano powder. This micrograph shows that spherical nano CdS particles get agglomerated to form a hierarchical flower like structure. These agglomerations can be further clearly visualized from FETEM micrograph as shown in fig. 2(b). The average particle size calculated by sampling 50 number of nanoparticles within the marked circular regions of fig. 2(a, b) are found to be 28 and 17 nm respectively.

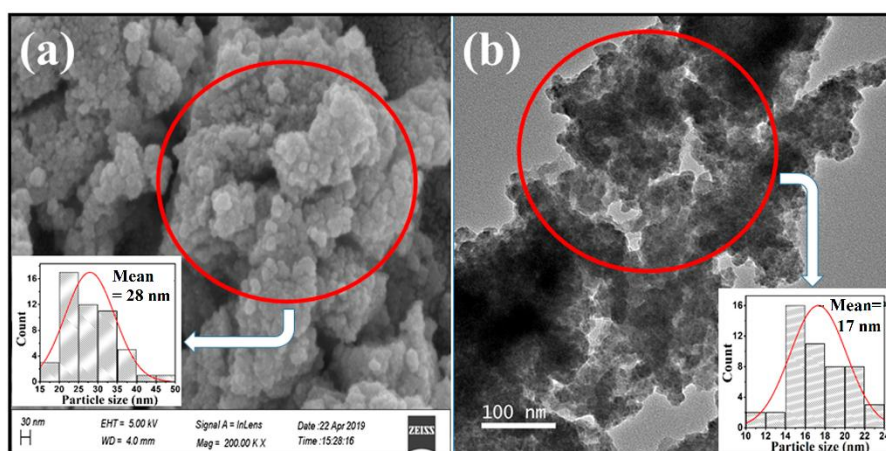
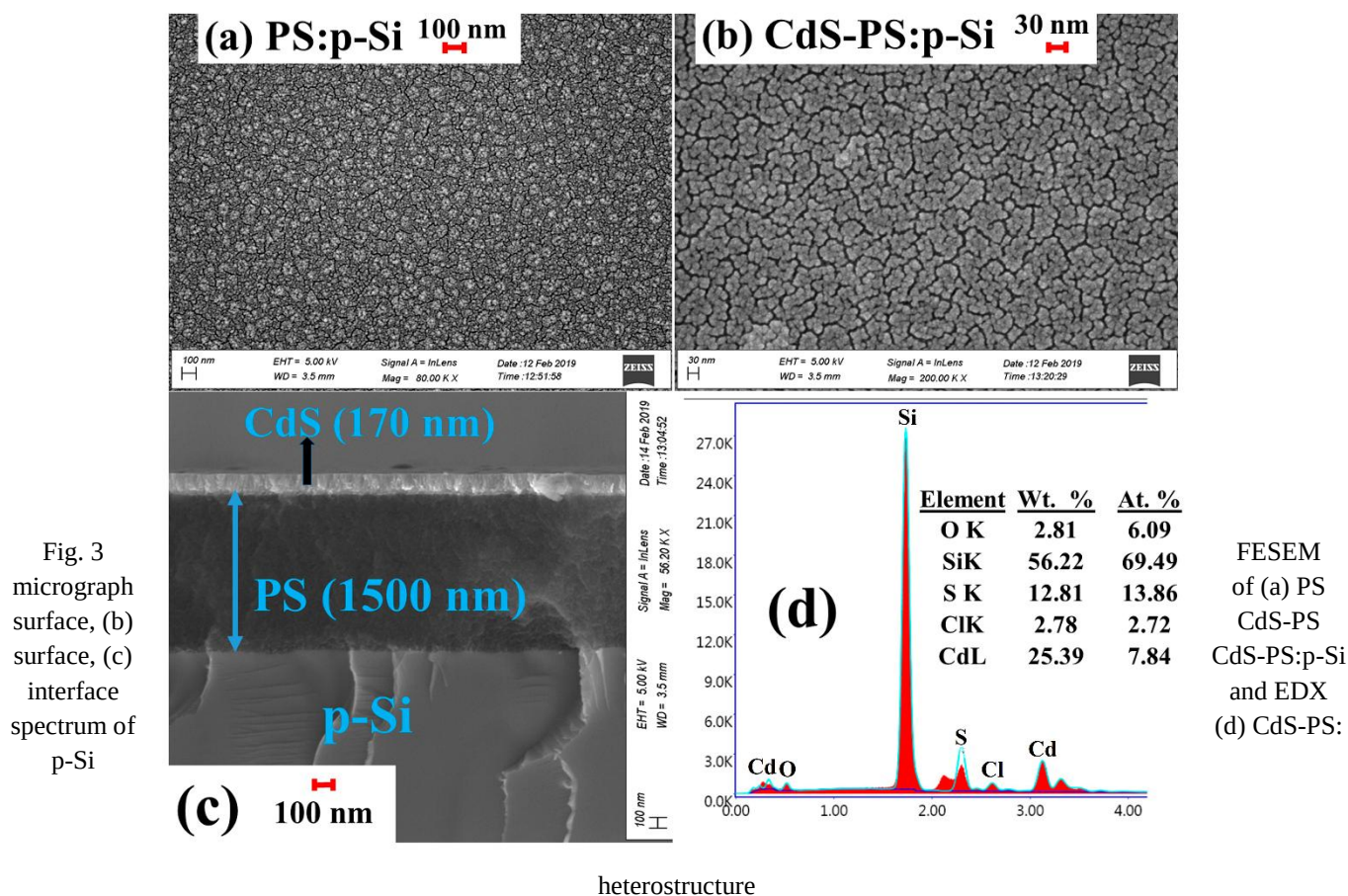


Fig. 2 (a) FESEM and (b) FETEM micrographs of CdS nano powder [insets- particle size histograms]

FESEM micrographs of prepared PS:p-Si and CdS-PS:p-Si heterostructure are shown in fig. 3. Fig. 3(a) shows top view surface morphology of PS layer grown on p-Si substrate. This shows random distribution of mesopores of the PS layer. Fig. 3(b) shows the top view of CdS-PS:p-



Si heterostructure, showing hierarchical flower like CdS nanostructures grown on PS:p-Si surface. The basic advantage of such hierarchical structure is to increase surface to volume ratio of the heterostructure, which efficiently increases its photo sensing property. To visualize the interface between CdS and PS:p-Si structure, cross sectional view of FESEM micrograph of the heterostructure is observed. This is shown in fig. 3(c), which shows uniform growth of CdS layer with thickness of ~170 nm on uniformly etched PS layer. To ascertain the presence of CdS on the PS:p-Si heterostructure, EDX analysis is carried out, which is presented in fig. 3(d). Presence of Cd is found to be 25.39 weight percent and 7.84 of atomicity percentage and for S it is 12.81% by weight and 13.86% by atomicity. Apart from Cd and S, other elements with atomicity (%) namely Si (69.49%), O (6.09%) and Cl (2.72%) are also present. Atomic weight of Si is maximum which is likely because CdS is deposited on PS:p-Si substrate. Presence of oxygen is attributed to growth of atmospheric oxygen on PS surface. The presence of chlorine (Cl) atom is a consequence of residual (CdCl_2), which is used during chemical synthesis of CdS.

3.2 Optical studies

Optical analysis of the prepared CdS and the heterostructures are observed via UV-visible absorbance, reflectance spectra and photoluminescence (PL) spectra. Fig. 4(a) shows UV-vis absorbance spectra of nanostructured CdS with onset and offset absorption band edge at ~482 and ~440 nm respectively, which correspond to the bandgap energy of ~2.6 eV as calculated using well known Tauc's equation and is depicted in fig. 4(b). Calculated bandgap energy of nanostructured CdS is appreciably larger than its bulk counterpart (2.42 eV). This blue shift of 0.18 eV for CdS nanostructure compared to its bulk one is

attributed to the quantum size effect in CdS nanostructure [19]. These facilitate employing such materials for photo detection in the shorter wavelength range of 250-490 nm of the UV-vis spectrum. Fig. 4(c) shows normalized UV-vis reflectance spectra of PS:p-Si and CdS-PS:p-Si heterostructure in the wavelength range of 200-800 nm. PS:p-Si structure exhibits multiple interference pattern due to Fabry-Perot interference of light reflected from various interfaces of PS:p-Si heterostructure [20]. However, on CdS deposition, i.e. for CdS-PS:p-Si heterostructure there is an increment as well as blue shift of interference patterns as compared to bare PS:p-Si structure. This can be attributed to the change in refractive index of the heterostructure on CdS deposition. Fig. 5(a-c) shows PL spectra of the prepared CdS, PS: p-Si and CdS-PS:p-Si heterostructure for excitation wavelength of 250 nm. Bare CdS nanostructure exhibits two broad emission bands within the spectral range of 430-490 nm and 510-600 nm respectively. On deconvoluting the peaks using Gaussian fitting analysis, peak positions are found to be at 450, 460, 468, 482 and 533 nm with FWHM of ~5, 23, 10, 8 and 22 nm respectively. The luminescence bands centred at ~ 450, 460, 468 and 482 nm are due to recombination of excitons or shallow trapped electron-hole

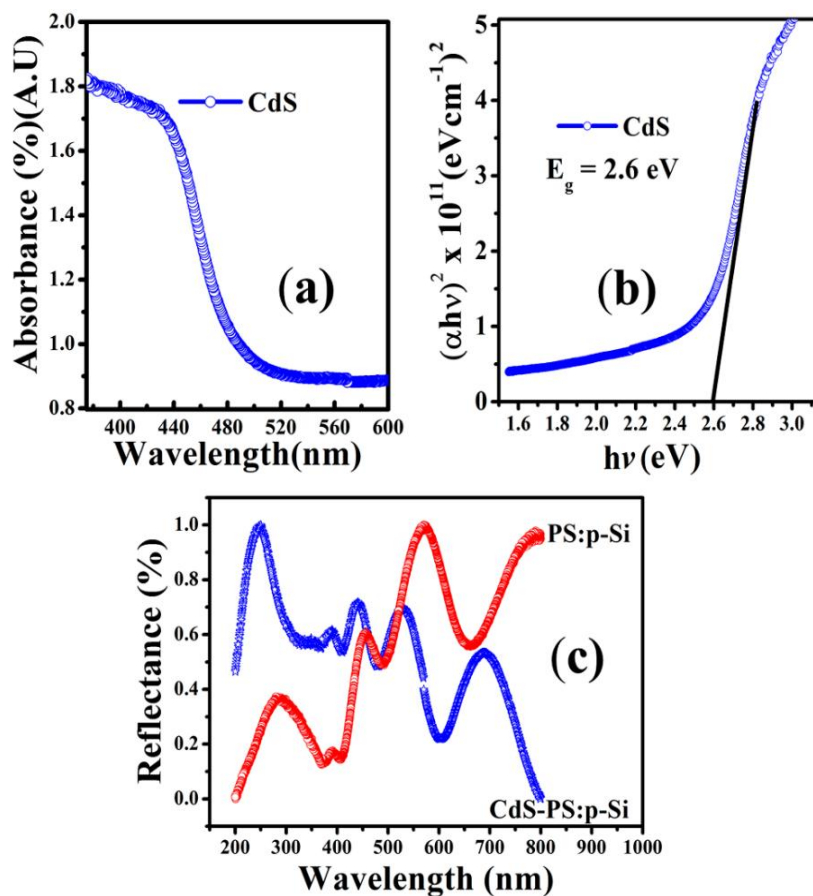


Fig. 4 (a) UV-vis absorbance spectra, (b) $(\alpha hv)^2$ vs. $h\nu$ plot of nanostructured CdS, (c) normalized UV-vis reflectance spectra of PS:p-Si and CdS-PS:p-Si heterostructure

pairs and are attributed to band edge emission of CdS nanoparticle [21, 22]. The green emission band centred at ~533 nm is a contribution of defect level electronic transition from sulphur vacancy (V_s) to the valence band of CdS nanostructure [23]. The luminescence mechanism for CdS nanostructure is shown in band representation in fig. 5(d). PS: p-Si heterostructure exhibits two distinct luminescence bands at ~468 and ~590 nm. The weak luminescence band at ~468 nm with FWHM of ~12 nm is attributed to

SiO₂ related transition [24]. The strong luminescence band at ~590 nm with FWHM of ~90 nm is attributed to band-to-band transition due to quantum confinement effect in Si nanostructure [25]. However, on deposition of CdS on PS: p-Si structure, it is observed that CdS related luminescence bands retain their original peak positions while PS: p-Si related luminescence band shows appreciable peak shifting [fig. 5(c)]. On peak deconvolution using similar Gaussian fitting for the spectral range of 430-490

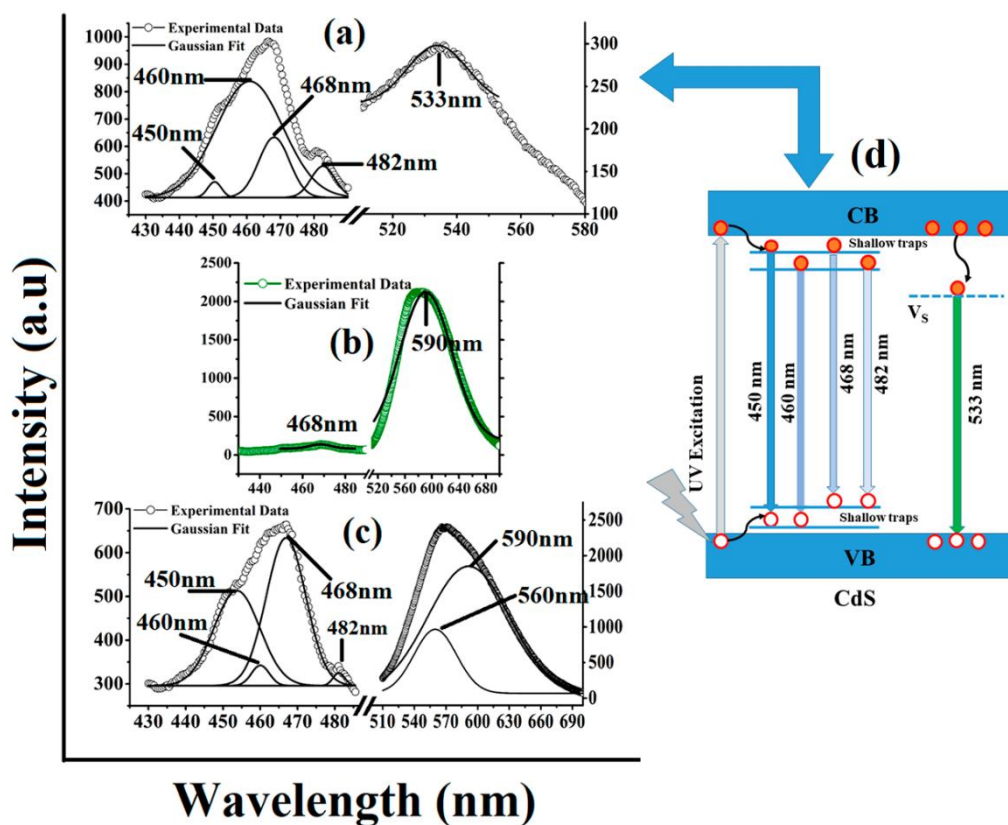
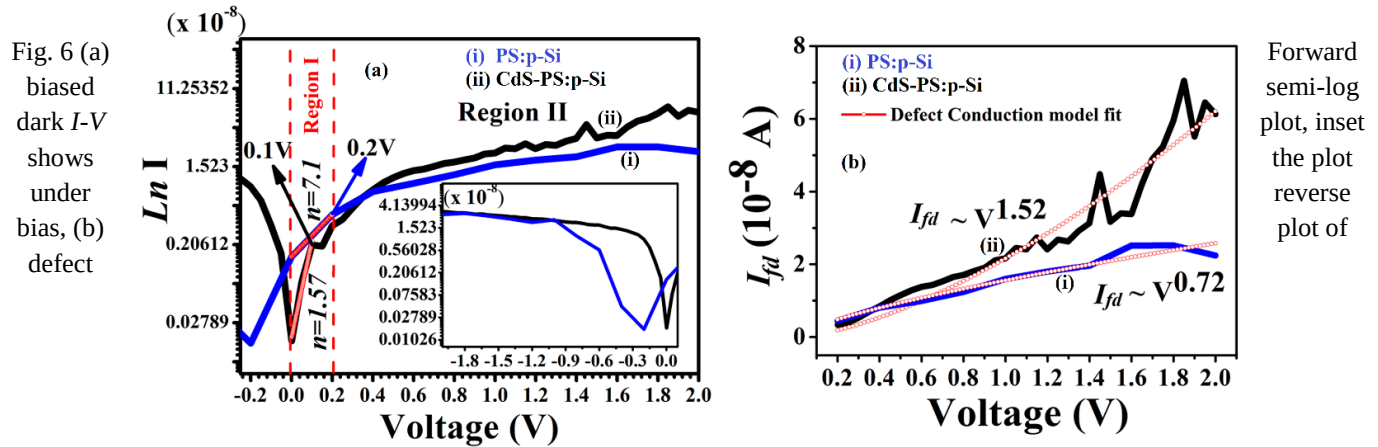


Fig. 5 PL spectra of (a) CdS nanostructure, (b) PS:p-Si structure, (c) CdS-PS:p-Si heterostructure and (d) schematic of electronic transitions involved in luminescence of CdS nanostructure

nm and 510-700 nm, it has been found that SiO₂ related luminescence band overlapped with 468 nm luminescence peak of CdS. Along with the characteristic PS:p-Si luminescence band at ~590 nm, a new defect related luminescence peak appeared at ~560 nm compensating the 533 nm peak of bare CdS nanostructure. The related 560 nm luminescence band is attributed to the presence of Cl atoms on sulphur vacancy sites of CdS nanostructure in CdS-PS: p-Si heterostructure [26]. Thus, from the PL studies it can be stated that in CdS-PS:p-Si heterostructure traps and defect states play crucial role for electron- hole pair recombination, which subsequently have a direct role on its electrical and photo sensing abilities to be discussed in next subsections.

3.3 Electrical and photo response studies

For efficient performance of any photo sensing device, it is highly required that a very low magnitude of dark current (I_d) flows through the device. To inspect this, current-voltage (I - V)



characteristics of both the devices are recorded for a voltage range of -2 V to +2 V under dark and illumination condition at room temperature. Fig. 6(a) shows forward biased *semi-log* I - V plot of dark currents for biasing upto 2V. The inset of the figure shows similar plot under reverse bias condition for voltage range up to -2V. The heterostructures produce low magnitude of reverse leakage current and are found to be ~32 nA at -2V for both the heterostructures. Also, decrease in the I_d value for CdS-PS: p-Si compared to PS: p-Si structure in low voltage region (-0.2 V to 0.2 V) is observed due to dominance of surface charge recombination in CdS-PS: p-Si junction compared to PS:p-Si junction. To know more about the electrical nature of the prepared heterostructures, forward biased I - V curve is divided into two regions. In region I (0-0.2V), current is seen to increase linearly up to 0.1V for CdS-PS: p-Si. This linearity is valid up to 0.2V for PS:p-Si structure. Such behaviour can be attributed to thermally generated e^-h^+ pairs at CdS-PS:p-Si and PS:p-Si junctions at room temperature. Applying thermionic emission model in this region, ideality factor (n) and saturation current (I_s) are calculated. The related equations [27] are mentioned as follows,

$$I = I_s \exp \frac{eV}{nkT} \quad (1)$$

$$n = \frac{e}{kT} \frac{dV}{d \ln I} \quad (2)$$

Where k is Boltzmann constant, e is electronic charge, T is room temperature. The values of the ideality factors (n) are calculated to be 1.57 and 7.1 and saturation current (I_s) as 0.2 and 1.5 nA respectively for the two devices. The values of n and I_s for the nanostructured CdS deposited on the heterostructure are seen to decrease suggesting improvement in device quality in comparison to the bare PS:p-Si structure. In region II (> 0.2V), forward current is seen to increase exponentially with applied voltage. Basically, in this region conduction takes place due to drift motion of charge carriers. To explore more about the conduction mechanism in this regime, defect conduction model is fitted to the forward I - V curve of both the devices using the equation $I_{fd} = CV^m$ [fig. 6(b)], where I_{fd} is forward dark current, C is the proportionality constant, V is the applied forward voltage and m is the fitting or scaling parameter. This model fits well for CdS-PS:p-Si device with adjusted *R-square* value of 0.94, giving m values of 1.54.

However, m values for bare PS:p-Si is found to be 0.72. So, it can be asserted that current conduction phenomenon in the former device is controlled by traps and defects present within the heterostructure [28].

To observe photoresponse characteristics, spectral responses of both the devices are recorded in the wavelength (λ) range of 250-500 nm at -2V bias. Responsivity (R_λ) and external quantum efficiency (EQE) vs. λ plots are shown in fig. 7. The equations 3.3 and 3.4 mentioned below are fin use to calculate these parameters.

$$R_\lambda = \frac{I_{ph}}{SP} \quad (3)$$

$$EQE = \frac{hc}{e\lambda} R_\lambda = \frac{1240 R_\lambda}{\lambda (nm)} \quad (4)$$

Where, I_{ph} is the photocurrent in ampere, S is effective device area in cm^2 , P is power density in Wcm^{-2} , λ is the wavelength of incident radiation in nm , e is the electronic charge, h is Planck's constant and c is the velocity of light. It is evident from fig. 7 that CdS deposited heterostructure exhibits enhanced maximum R_λ and EQE at 400 nm with the values of $\sim 0.6 AW^{-1}$ and $\sim 180\%$ compared to $\sim 0.3 AW^{-1}$ and $\sim 85\%$ for the bare PS: p-Si structure. Along with this, one can notice an additional enhanced R_λ peak overlapping the weak PS:p-Si peak at ~ 440 nm. This particular peak coincides with the band edge absorbance peak of nano CdS as shown in the inset of the figure. The high EQE value ($> 100\%$) is believed to be due to the presence of trapped charge carriers in CdS-PS heterostructure, which upon illumination results in reduction of Schottky barrier height [29, 30]. Since, R_λ and EQE describe the performance figures of merit of any photo sensing heterostructure, 100% enhancement in respective values ensure that CdS-PS:p-Si device performance is far superior as compared to the bare PS:p-Si one. On application of high external bias of -2V dissociation of photo-generated e^-h^+ pairs occur and are transported and collected by the Al electrodes before recombination, thereby reducing the charge carrier recombination. Fig. 8(a) shows linear I - V plot of CdS-PS: p-Si heterostructure under incident radiation wavelength of 400 nm. From the plot it is evident that the heterostructure shows almost symmetric enhancement of rectification in forward and reverse bias condition with a photo to dark current ratio (I_{ph}/I_d) of 3.15 and 10.26 at $\pm 2V$ respectively. The symmetrical rectification enhancement under illumination is basically due to the lateral MSM configuration of the device [31]. The photoresponse is recorded by changing

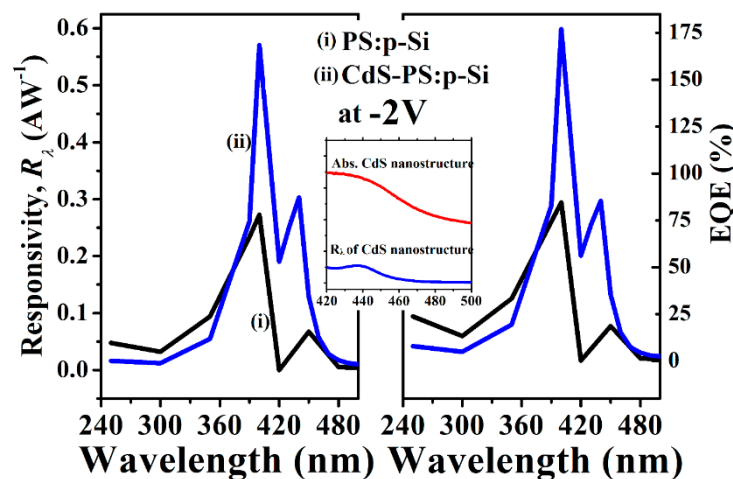


Fig. 7 plot of R_λ and EQE

vs. λ at -2V for both the

heterostructure PDs, inset shows absorbance and R_λ spectra of nanostructured CdS

irradiation intensities ($33-71\mu W cm^{-2}$) at a fixed 400 nm wavelength radiation of light. The intensity dependent I_{ph} graph at -2V is shown in fig. 8(b). This shows power law dependence, i.e. I_{ph} follows

power law equation $I_{ph} = AP^\alpha$ where A is proportionality constant, P is intensity of the irradiating wavelength and α is scaling parameters whose specific value reveals the type of carrier dynamics present [32]. $\alpha = 1$ implies number density of trapped e^- (N_{trap}) $\gg$ number density of free e^- (N), which is first order carrier dynamics. $\alpha = 0.5$ implies $N_{trap} \ll N$, which is for second order carrier dynamics [33]. In the present work, α value is 0.8, which suggests carrier transport in CdS-PS:p-Si heterostructure to be governed by the combination of first and second order dynamics. The on-off transient kinetics of the heterostructure under different irradiation intensities at bias voltage of -2V with on-off time interval of 10 s is shown in fig. 8(c). From the curve it is clear that the heterostructure shows steep rise in current when the incident light is turned on and remain in a stable state until the light is turned off and relaxes to its initial state. Also, with increasing irradiation intensity from 33 to 71 μWcm^{-2} , the on-off ratio increases from 1.32 to 2.92, indicating strong influence of irradiation intensity on its switching behaviour. Since the nature of curve in fig. 8(c) represents sigmoidal growth in the ‘ON’ state and exponential decay in the ‘OFF’ state, response and recovery times of the heterostructure can be calculated using the following fitting equations.

$$I = \frac{A_1 - A_2}{1 + e^{(t-t_0)/\tau}} + A_2 \quad (5)$$

$$I = I_0 + Ae^{-t/\delta} \quad (6)$$

Equation (3.5) represents Boltzmann function, where A_1 , A_2 and τ are onset value, offset value and response time constant respectively. Equation (3.6) represents exponential decay function, where I_0 , A and δ are offset value, amplitude and recovery time constant respectively. From the fitted curve as shown in fig. 8(d), the value of τ and δ are found to be 0.16 ± 0.03 and 0.35 sec

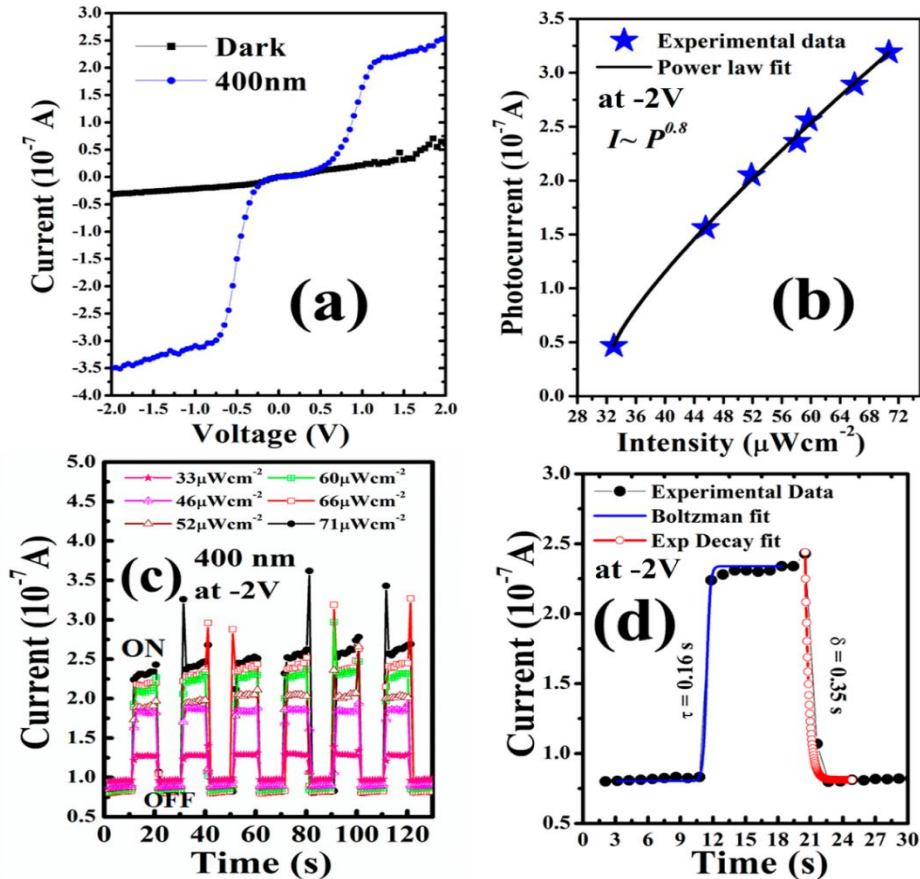


Fig. 8 (a) Linear I - V plot under dark and illumination condition, (b) intensity dependent I_{ph} plot, (c) On-off kinetics under different intensity of 400 nm and (d) response and recovery time calculation for CdS PS:p-Si heterostructure PD at -2V

respectively. Recovery time is longer than the response time, as thermal relaxation of e^- is a slower process compared to the photoexcitation of e^- [34, 35]. The response time is quite high compared to some of the earlier reported high-speed PDs [36-38], which can be attributed to the presence of interfacial defects within the heterostructure. Further improvement towards the response time can be done by optimizing the thickness of CdS and PS layer and also by annealing the PS prior to electrode deposition. On the basis of performance parameters, Al: CdS-PS: p-Si is compared with some of the best reported silicon-based hybrid heterostructure PDs and the results are summarized in Table 1. From the table it is evident that performance of the present heterostructure PD is more or less at par with some devices [6, 17, 36-44], but in

Table 1 Comparative performance analysis of Al: CdS- PS:p-Si heterostructure PD with some of the best reported Si-based

PDs in the literature.

Device	Measurement condition [wavelength (λ), intensity (I)]	Reverse saturation current (nA)	Responsivity (AW^{-1})	Response time (ms)		Ref.
				Rise (τ)	Decay(δ)	
CuO: Si (p-n)	$\lambda = 405$ nm $I = 0.48$ Wcm^{-2}	—	3.89×10^{-4}	0.06	0.08	[36]
Graphene: Si	$\lambda = 488$ nm $I = 0.04$ μWcm^{-2}	3×10^2	0.23	5	5	[38]
MoS ₂ : Si heterojunction	$\lambda = 808$ nm $I = 30$ $mWcm^{-2}$	—	0.30	3×10^{-3}	4×10^{-2}	[37]
MoS ₂ : Si heterojunction	$\lambda = 514$ nm $I = 0.8$ $mWcm^{-2}$	10	0.47	10	10	[39]
PS- MSM device	$\lambda = 530$ nm $I = 1.5$ $mWcm^{-2}$	—	0.08	510	450	[42]
n-SiCN/p-PS/p-Si	$\lambda = 254$ nm $I = 0.5$ $mWcm^{-2}$	—	—	7×10^3	1×10^4	[43]
PS: n-Si MSM device	$\lambda = 450$ nm $I = 259$ μWcm^{-2}	95	2.00	1000	1000	[41]
	$\lambda = 250$ nm $I = 28.2$ μWcm^{-2}		1.42	700	800	
p-Si/n-CdS core shell nanowire	$\lambda = 700$ nm	—	1.37	—	—	[17]
CdS:PS	$\lambda = 577$ nm	—	1.7	—	—	[6]
Al:CdS-PS:p-Si	$\lambda = 400$ nm $I = 71$ μWcm^{-2}	32	0.60	160	350	Present work [19]

some other cases it's even superior in performance in spite of simpler and less expensive fabrication technique utilised.

4. Conclusion

MSM device based on nanostructured CdS-PS: p-Si heterostructure has been successfully developed with simpler inexpensive technique. Light sensing abilities in a UV-vis spectral range between 250 to 500 nm is tested for this device. FESEM and FETEM analyses confirm formation of CdS nanostructure. Cross sectional FESEM view confirms uniform interface between CdS and PS: p-Si structure. EDX analysis confirms presence of Cd and S in the prepared heterostructure which is also supported by UV-vis reflectance measurement showing increment as well as blue-shift of the interference pattern on CdS deposition. PL study reveals trap and defect assisted luminescence bands at 450, 460, 468, 482 and 533 nm for nanostructured CdS along with characteristic band gap emission of PS at ~590 nm. These traps and defects in turn control electrical current conduction phenomena within the heterostructure as confirmed from dark current analysis. The heterostructured PD exhibits low leakage current and under illumination of 400 nm radiation shows high responsivity, high EQE and fast response time of ~160 ms at -2V. All these facts assure the device to be applicable as an eligible candidate for UV-vis photodetector and switching device.

References

1. X. Wang et al., *Adv. Funct. Mater.* **23**, 1202 (2013).
2. F. Sarubbi et al., in *Proceedings of 38th European Solid-State Device Research Conference* (2008), pp. 278-281.
3. G.F. Knoll, *Radiation detection and measurement, 4th edition* (Wiley, 2010), pp. 265-302.
4. H.A. Atwater and A. Polman, *Nat. Mater.* **9**, 205 (2010).
5. H. Chen et al., *Mater. Today* **18**, 493 (2015).
6. T. Zhou et al., *Nano Energy* **50**, 118-125 (2018).
7. C. Li et al., *Solar Energy Mater. Solar Cells* **172**, 341 (2017).
8. J. Wei et al., *Opt. Lett.* **42**, 911 (2017).
9. L.T. Canham et al., *Appl. Surf. Sci.* **102**, 436 (1996).
10. Y. Qu et al., *Nano Lett.* **9**, 4539 (2009).
11. X. Li and P.W. Bohn, *Appl. Phys. Lett.* **77**, 2572 (2000).
12. L. U. Khan and M.J.M. Jaffar, *J. Mater. Sci.: Mater. Electron.* **28**, 7677 (2017).
13. D. Max Roundhill and J.P. Fackler Jr., *Optoelectronic Properties of Inorganic Compounds* (Springer, 2013).
14. F. Meng and Z. Sun, *Appl. Surf. Sci.* **255**, 6715 (2009).
15. E.M. Kaidashev et al., *Appl. Phys. Lett.* **82**, 3901 (2003).
16. R.A. Ismail, K.S. Khashan and A.M. Alwan, *Silicon* **9**, 321 (2017).
17. S. Manna et al., *J. Phys. Chem. C* **116**, 7126 (2012).
18. N.D. Arora, J.R. Hauser and D.J. Roulston, *IEEE Trans. Electron Dev.* **29**, 292 (1982).
19. M. Das, S. Sarmah and D. Sarkar, *J. Mater. Sci.: Mater. Electron.* **30**, 11239 (2019).
20. A.I. Olivaa et al., *Thin Solid Films* **391**, 28 (2001).
21. C. Pacholski et al., *J. Am. Chem. Soc.* **127**, 11636 (2005).
22. L. Spanhel and M.A. Anderson, *J. Am. Chem. Soc.* **112**, 2278 (1990).
23. Q. Wang et al., *Sci. Rep.* **5**, 13593 (2015).
24. R.M. Nirmal, K. Pandian and K. Sivakumar, *Appl. Surf. Sci.* **257**, 2745 (2011).
25. H.Z. Song and X.M. Bao, *Phys. Rev. B* **55**, 6988 (1997).

26. J.L. Gole et al., Phys. Rev. B **56**, 2137 (1997).
27. N. Maticiuc et al., Thin Solid Films **535**, 184 (2013).
28. S.M. Sze, *Physics of Semiconductor Devices*, 2nd edition (Wiley, 1981) pp. 849.
29. A. Chouket, et al., J. Appl. Phys. **114**, 243105 (2013).
30. A.A. Hussain et al., Appl. Mater. Interfaces **8**, 4258 (2016).
31. D.S. Tsai et al., ACS Nano **7**, 3905 (2013).
32. P.R. Berger, in *Proceedings of SPIE* (2001), Vol. 4285.
33. L.J. Willis et al., Nano Lett. **9**, 4191 (2009).
34. R.H. Bube, *Photoconductivity of Solids* (Wiley, 1960).
35. E. Baek et al., Nano Res. **8**, 1229 (2015).
36. G.W. Robinson and R.P. Frosch, J. Chem. Phys. **38**, 1187 (1963).
37. Q. Hong et al., Appl. Mater. Interfaces **6**, 20887 (2014).
38. L. Wang et al., Adv. Funct. Mater. **25**, 2910 (2015).
39. X. An et al., Nano Lett. **13**, 909 (2013).
40. S. Mukherjee et al., Nanotechnology **28**, 135203 (2017).
41. C.D. Wessendorf et al., Sol. RRL **2**, 1800056 (2018).
42. M. Das, S. Sarmah and D. Sarkar, Superlatt. Microstruct. **101**, 228 (2017).
43. A.A. Thahe et al., Optik **168**, 424 (2018).
44. T.H. Chou et al., Sensors Actuators, A Phys. **279**, 462 (2018).